AIR UNIVERSITY

Manufacturing Automation

Complex Engineering Activity (CEA)

Course Code: MT-451 L
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Abstract

This report presents the design, simulation, and implementation of a Complex Engineering Activity (CEA) for the Manufacturing Automation course. The activity required the analysis and development of PLC-based automation solutions for three real-world industrial problems, including a traffic light control system, a batch processing system, and a conveyor-based rolling system. The project followed a structured engineering approach, starting from process understanding and logical breakdown, followed by control strategy development and ladder logic programming. Industrial automation principles such as sequencing, safety interlocks, timing control, and operator interaction were emphasized. The designed solutions were simulated and verified to ensure correct, safe, and reliable operation under varying conditions. The activity is mapped to relevant Course Learning Outcomes (CLOs) and Program Learning Outcomes (PLOs), demonstrating competency in industrial automation, PLC programming, and organized engineering work.

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1. Introduction

Manufacturing automation plays a vital role in modern industrial systems by improving productivity, safety, and consistency while reducing human error. Programmable Logic Controllers (PLCs) are widely used in industrial environments due to their robustness, flexibility, and real-time control capabilities. This Complex Engineering Activity (CEA) was assigned to provide hands-on experience in analyzing, designing, and implementing PLC-based automation solutions for realistic industrial scenarios.

The CEA involved three automation problems requiring logical sequencing, safety considerations, and operator-friendly design. Students were required to understand each process, identify inputs and outputs, develop flowcharts, and implement ladder logic programs using standard PLC instructions. The final solutions were simulated and verified to ensure reliable operation.

2. Project Objectives

The main objectives of this Complex Engineering Activity are as follows:

• To analyze real-world industrial automation problems and identify required inputs, outputs, sensors, and actuators.
• To design structured control strategies using flowcharts and logical sequencing.
• To develop PLC ladder logic programs using timers, counters, latches, and sequencers.
• To simulate and verify PLC programs for correct and safe operation.
• To demonstrate alignment with Course Learning Outcomes (CLOs) and Program Learning Outcomes (PLOs).

3. Problem Overview and System Requirements

3.1 Overview of Assigned Automation Problems

This CEA consists of three manufacturing automation problems:

1. **Traffic Light Control System:** A PLC-based traffic light controller using a sequencer to manage vehicle and pedestrian signals safely under low-visibility and peak traffic conditions.
2. **Batch Processing System:** An automated syrup production system capable of operating in both single-batch and multiple-batch modes using counters, sensors, heating, and mixing control.
3. **Conveyor and Rolling System:** A textile conveyor system with three selectable operating modes, ensuring safe and efficient handling of fabric rolls.

3.2 Functional Requirements

The functional requirements for the systems include:

• Sequential control of processes using PLC logic
• Proper synchronization between different system components
• Operator control through Start/Stop switches and mode selectors
• Accurate timing, counting, and sequencing
• Visual indication of system status

3.3 Safety and Operational Constraints

Safety considerations were integrated into all designs, including:

• Immediate system stop on Stop or Emergency input
• Prevention of unsafe operation (e.g., rolling without a roll present)
• Adequate clearance and delay times to avoid collisions or hazards
• Operator-friendly logic without the need for continuous button pressing

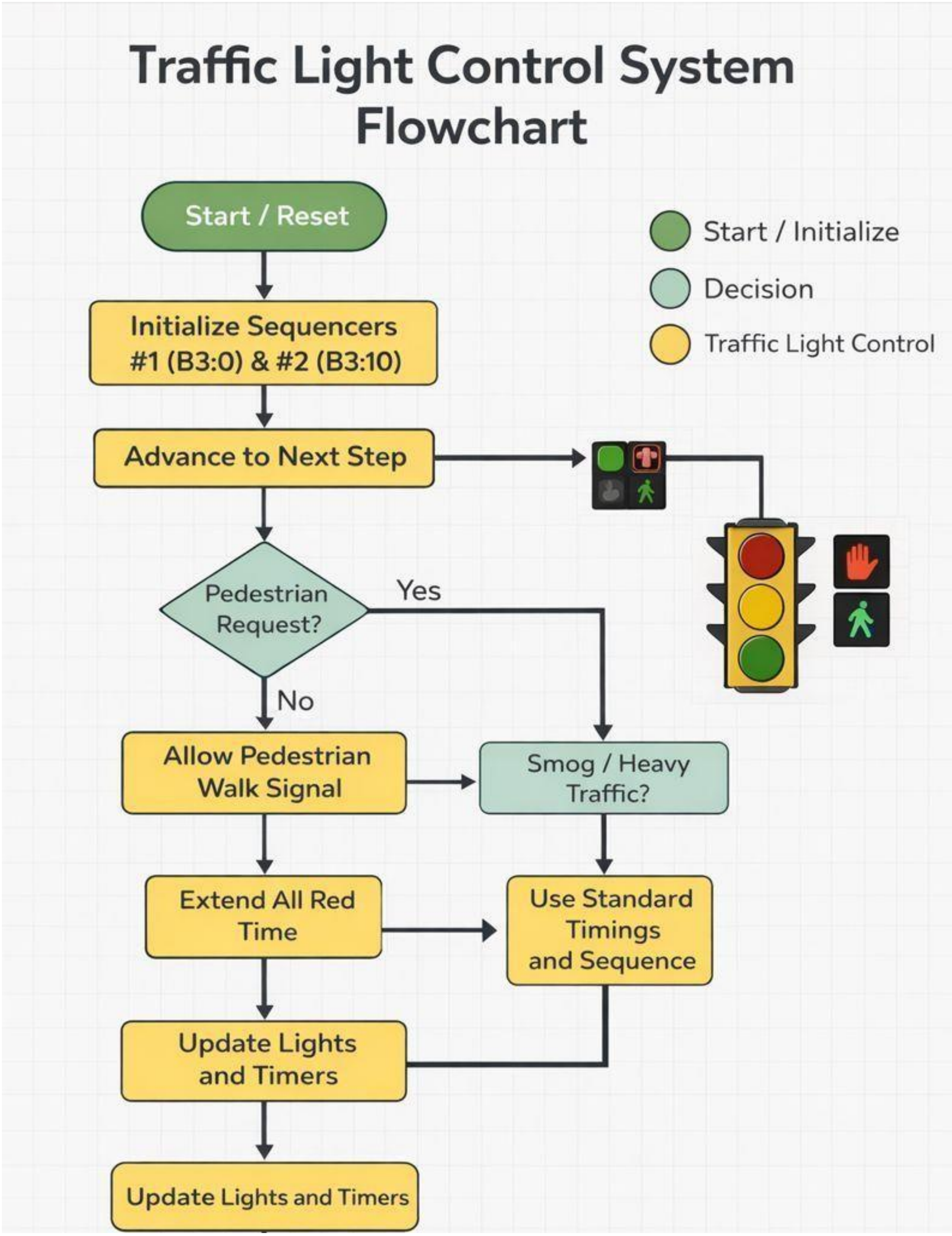
4. Methodology

4.1 Process Understanding and Logical Breakdown

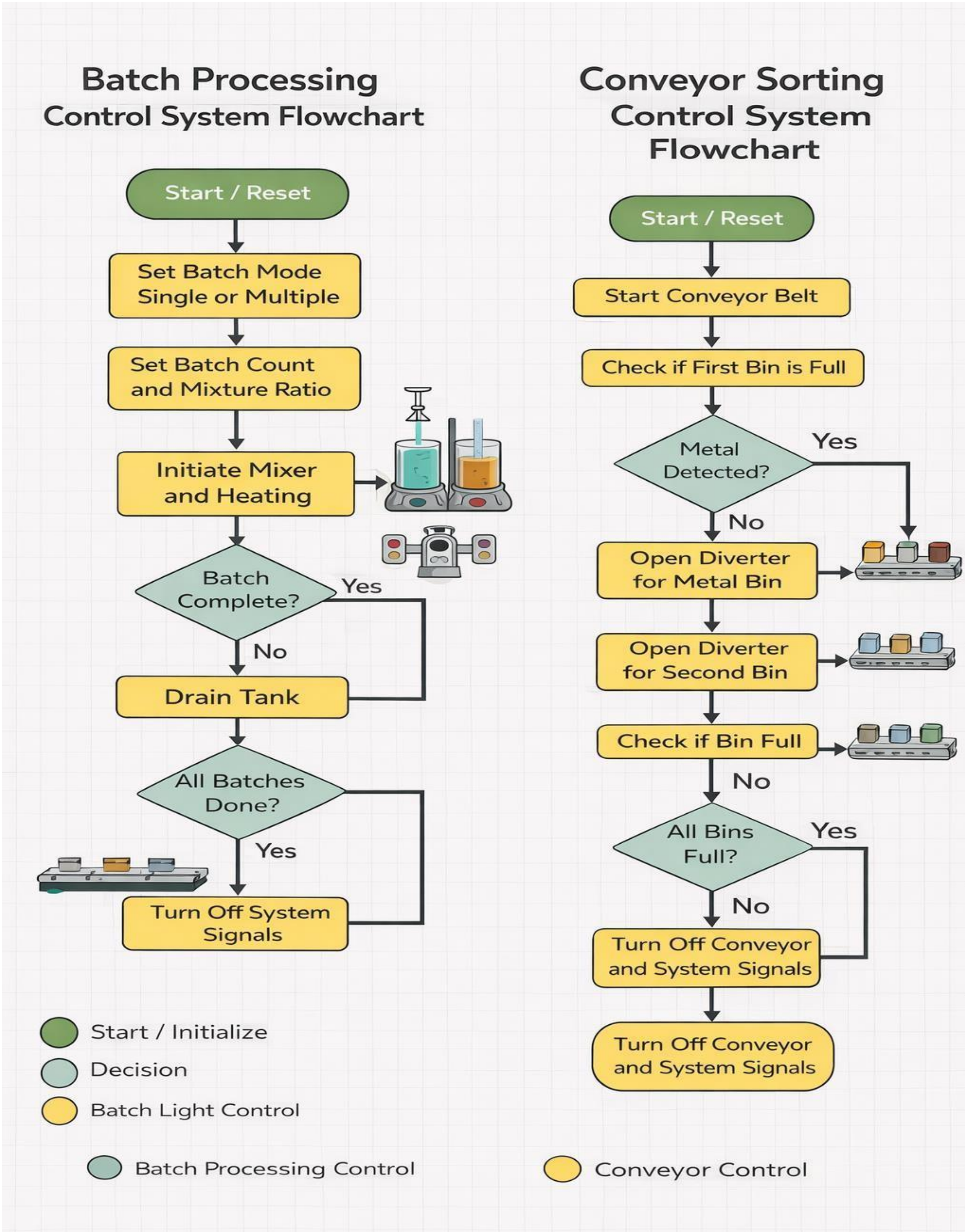
Each problem was first analyzed to understand the physical process and operational requirements. The complete process was logically broken down into sequential steps, decision points, and safety conditions. Inputs, outputs, and internal memory bits were identified during this phase.

4.2 Flowcharts

Flowcharts were developed to visually represent the control logic, sequence of operations, and decision-making process for each system.



- Flow Chart of Traffic Light Control System



- Flow Charts of Batch Processing Control System and Conveyor Sorting Control System

5. PLC Control Strategy

5.1 Control Philosophy

The control philosophy focuses on safe, sequential, and reliable operation. Each system follows a clearly defined sequence, ensuring correct timing, interlocks, and operator control.

5.2 Ladder Logic Design

The ladder logic programs were designed using modular rungs, clear labeling, and standard PLC programming practices to improve readability and troubleshooting.

5.3 Use of PLC Instructions

The following PLC instructions were used across the automation problems:

• Timers (TON) for delays and flashing signals
• Counters (CTU) for batch counting and quantity control
• Latches and unlatches for state retention
• Sequencers (SQO) for traffic light operation

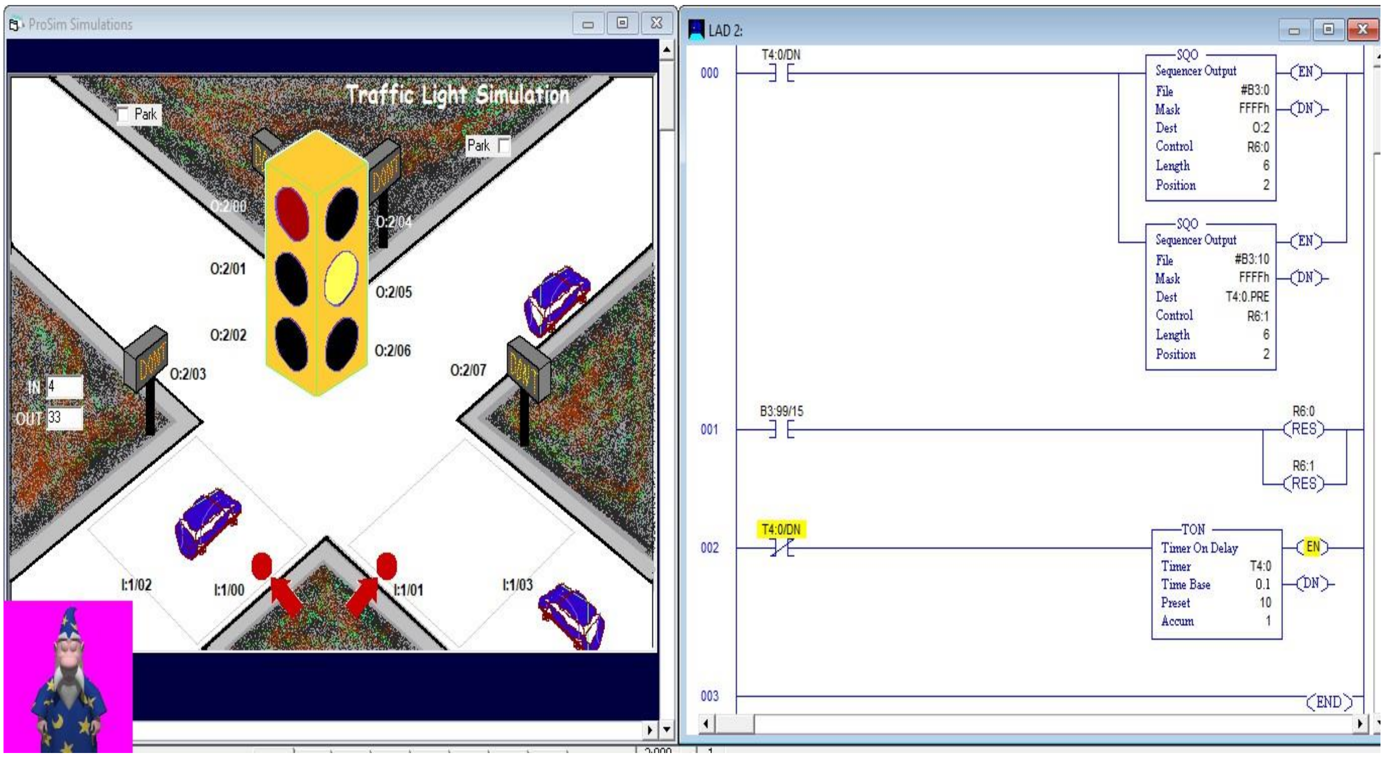
6. Implementation Details

6.1 Problem 1: Traffic Light Control System

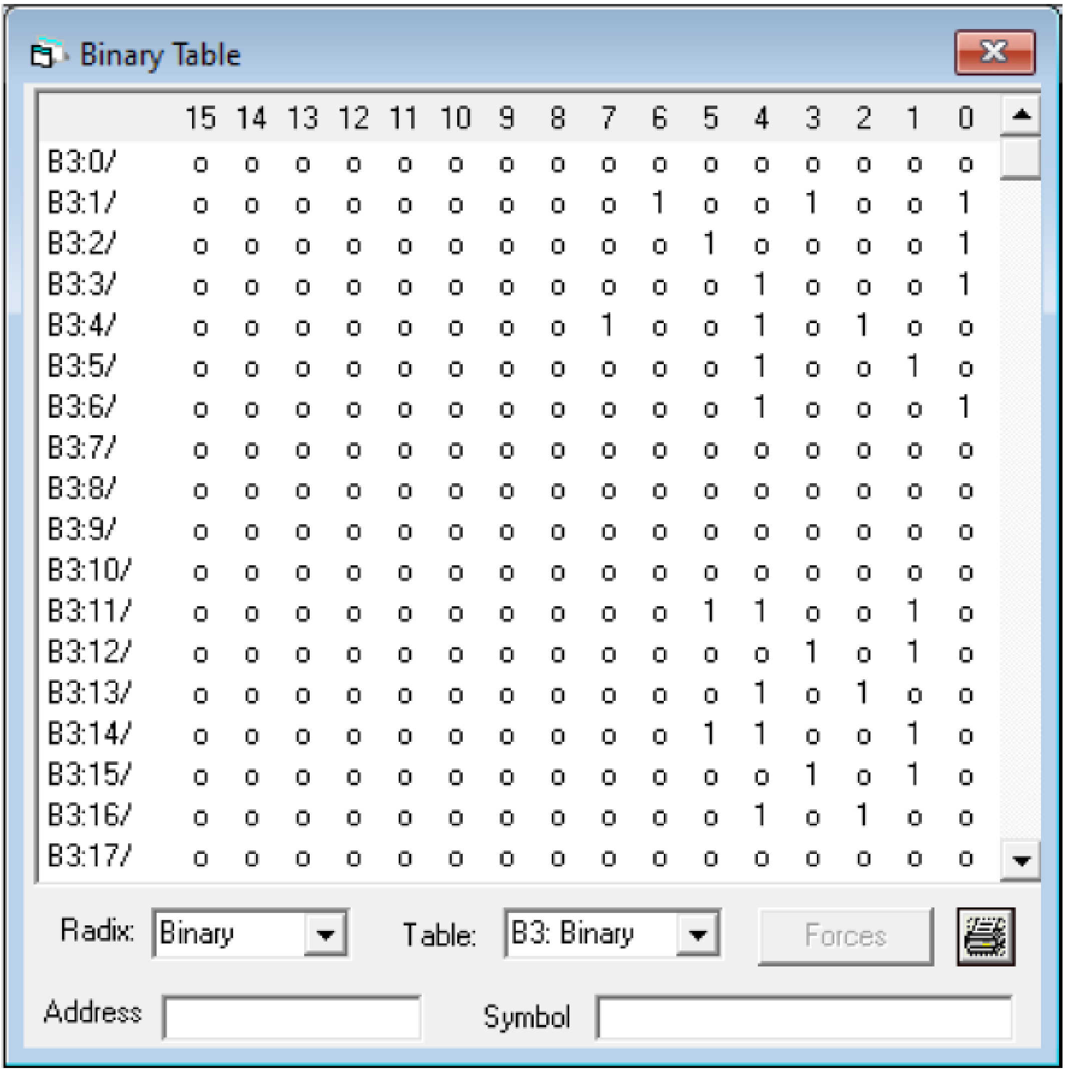
The traffic light control system was implemented using a PLC-based sequencer approach. Key implementation details extracted from the simulation and ladder logic are summarized below:

- ❖ **Control Method** - PLC programming and simulation performed using **TLP LogixPro Simulator** (Traffic Light ProSim). - Traffic light operation implemented using **Sequencer Output (SQO)** instructions.
- ❖ **Sequencer Configuration** - **SQO 1 (Traffic Outputs):** - File: B3:0 - Mask: FFFFh - Destination: O:2 - Control Register: R6:0 - Sequence Length: 6
- ❖ **SQO 2 (Timing Control):**
  - o File: B3:10
  - o Mask: FFFFh
  - o Destination: T4:0.PRE
  - o Control Register: R6:1
  - o Sequence Length: 6
- ❖ **Timer Details** - Timer Type: **TON (Timer ON Delay)** - Timer Address: T4:0 - Time Base: 0.1 s - Preset values loaded by SQO (observed values: 50, 10) - Sequencer advances using T4:0/DN

- ❖ **Traffic and Pedestrian Logic** - Vehicle red, yellow, and green lights controlled through output module O:2. - Pedestrian Walk signals enabled during vehicle green phase. - Walk signal flashes during amber phase using timer-based logic.
- ❖ **Reset Logic** - Internal bit B3:99/15 used as master reset. - Activation resets sequencer control registers R6:0 and R6:1



- ▪ Ladder logic of traffic light control system



	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
B3:0/	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B3:1/	0	0	0	0	0	0	0	0	0	1	0	0	1	0	0	1
B3:2/	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	1
B3:3/	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1
B3:4/	0	0	0	0	0	0	0	0	1	0	0	1	0	1	0	0
B3:5/	0	0	0	0	0	0	0	0	0	0	0	1	0	0	1	0
B3:6/	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	1
B3:7/	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B3:8/	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B3:9/	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B3:10/	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
B3:11/	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0
B3:12/	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0
B3:13/	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
B3:14/	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0
B3:15/	0	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0
B3:16/	0	0	0	0	0	0	0	0	0	0	0	1	0	1	0	0
B3:17/	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

- Binary data table of SQO's

6.2 Problem 2: Batch Processing System

The batch processing system supports both single-batch and multiple-batch modes. Counters were used to measure product quantity, level sensors controlled filling and draining, and timers ensured proper heating and mixing durations. LED indicators were used to display batch completion and current batch number.\

❖ **Control Method**

- PLC programming implemented using **Ladder Logic** in **TLP LogixPro Simulator (Batch Mix Simulator)**.
- Sequential control achieved using internal bits, counters, timers, and sensor feedback.
- The process is fully automated once started, minimizing operator intervention.

❖ Operating Modes Implementation

Single Batch Mode

- Operator initiates the process by pressing the **Start push button**.
- **Pump 1** fills the tank with Product 1 while a counter tracks the exact quantity added.
- When the counter reaches its preset value, Pump 1 stops automatically.
- **Pump 2** then fills the tank until the **High-Level sensor** is activated.
- Upon full tank detection:
  - **Full Tank indicator light** turns ON.
  - **Heater** and **Mixer** start automatically.
- Heater continues operation until the **Thermostat** confirms the required temperature.
- Mixer remains ON during heating and continues for an additional **4 seconds** after the desired temperature is reached to ensure proper blending.
- **Pump 3** starts draining the tank after mixing completes.
- Pump 3 stops automatically when the **Low-Level sensor** indicates the tank is empty.
- The system resets and is ready for the next batch.

❖ Multiple Batch Mode

- Operator selects the number of batches using a **thumbwheel switch** (e.g., 5 batches).
- Percentage of Product 1 is also set to control the mixture ratio.
- After pressing Start, the PLC executes batch cycles automatically without further operator input.
- Each batch completes the full sequence: filling, heating, mixing, and draining.
- Upon completion of each batch:
  - The batch counter increments.
  - Corresponding **Batch Completion LED** turns ON (LED 1 for Batch 1, LED 2 for Batch 2, etc.).
- The current batch number is displayed using LEDs for real-time progress monitoring.
- Process stops automatically after the final batch is completed.

❖ Input Devices and Sensors

- **Start Push Button (I:1/0)** – Initiates batch operation.
- **High-Level Sensor (I:1/4)** – Detects full tank condition.
- **Low-Level Sensor (I:1/3)** – Detects empty tank condition.
- **Thermostat Input (I:1/2)** – Confirms target temperature reached.
- **Thumbwheel Switch** – Sets number of batches and mixture ratio.

❖ Counters and Timers

Counter Details

- **Counter Type**: CTU (Count Up)
- **Counter Address**: C5:0

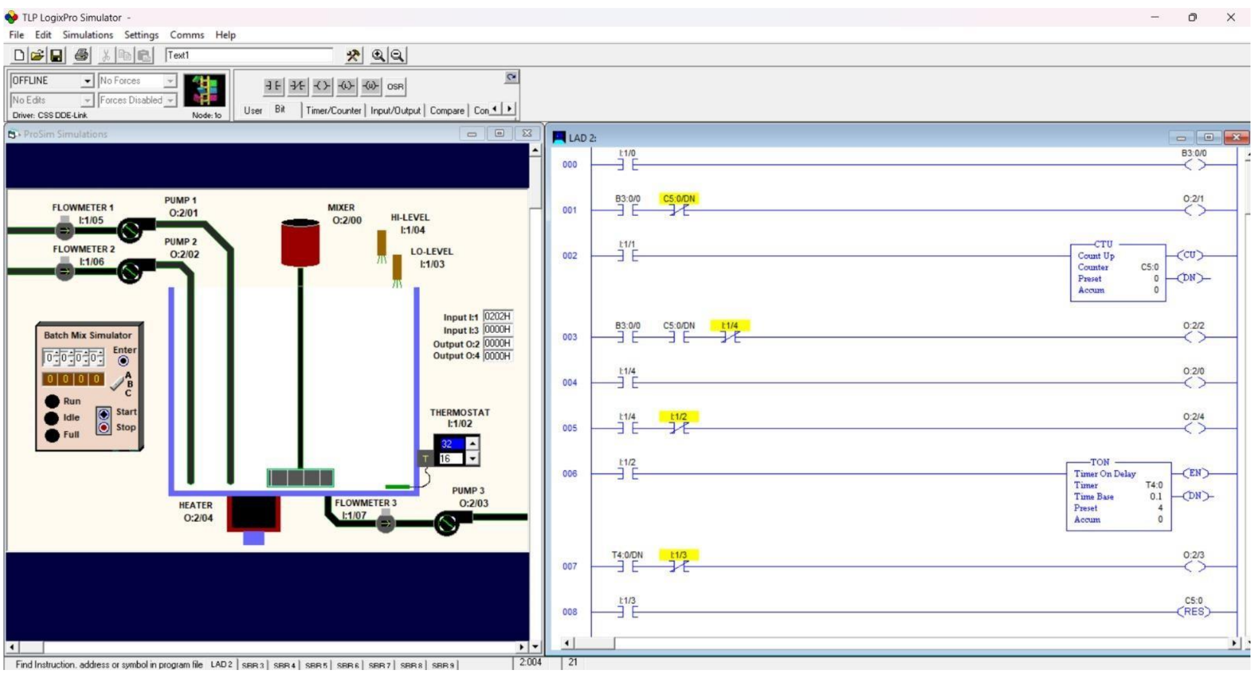
- **Preset Value**: Set according to required Product 1 quantity or batch count.
- **Usage**:
    - Measures Product 1 volume during filling.
    - Tracks number of completed batches in Multiple Batch mode.

Timer Details

- **Timer Type**: TON (Timer ON Delay)
- **Timer Address**: T4:0
- **Time Base**: 1.0 s
- **Preset Value**: 4 seconds
- **Usage**:
    - Ensures mixer runs for an additional 4 seconds after temperature is reached.

❖ **Output Devices**

- **Pump 1 (O:2/1)** – Fills Product 1 into the tank.
- **Pump 2 (O:2/2)** – Fills remaining liquid until tank is full.
- **Pump 3 (O:2/3)** – Drains the tank after mixing.
- **Heater (O:2/4)** – Raises syrup temperature.
- **Mixer (O:2/0)** – Mixes syrup during heating and timed delay.
- **Indicator LEDs** – Display Full Tank status, batch completion, and current batch number.



- Ladder logic of Batch Processing System

6.3 Problem 3: Conveyor and Rolling System

The conveyor system operates in three modes: continuous rolling, manual restart, and roll bypass. Sensors detect roll presence and completion, while operator inputs control system start and stop. Safety logic ensures no rolling occurs without a roll present.

❖ Input Devices and Sensors

- **Start Push Button**: Momentary input used for system start and step-wise control in Manual Restart mode.
- **Stop Push Button**: Master stop; immediately de-energizes all outputs and resets motion.
- **Roll Position Sensor**: Detects presence of a roll at the rolling machine.
- **Roll Complete Sensor / Timer Done Bit**: Indicates completion of rolling operation.
- **Mode Selector Inputs**: Internal bits used to select Mode A, Mode B, or Mode C.

❖ Output Devices

- **Conveyor Motors (A, B, C)**: Controlled through output coils for material transport.
- **Rolling Motor**: Enabled only when a roll is present and mode logic permits rolling.
- **Indicator Lights**:
  - RUN Light – System active
  - ROLLING Light – Rolling motor energized
  - DONE Light – Rolling operation completed

❖ Timer Details

- **Timer Type**: TON (Timer ON Delay)
- **Timer Address**: T4:0 (as observed in ladder logic)
- **Time Base**: 0.1 s
- **Preset Value**: 40 (used to define rolling duration)
- **Usage**:
  - Controls rolling duration.
  - Timer Done (DN) bit used to stop the rolling motor and trigger conveyor advancement.

❖ Safety and Interlock Logic

- Stop push button immediately disables all conveyors and motors.
- Rolling motor cannot start unless a roll is detected at the rolling position.
- Conveyor and rolling outputs are mutually interlocked to prevent unsafe operation.
- Operator-friendly design eliminates continuous button holding by using pulse-based control and internal latches.

❖ Reset and System Control Logic

- Stop condition resets internal mode and operation bits.

- Timer and rolling status bits are cleared upon system stop or cycle completion.
- End-of-program (END) instruction ensures proper PLC scan completion.

- Ladder logic of conveyor and rolling system

10. Team Members

The project tasks were distributed among team members to ensure efficient completion.

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- Saad Bin Awais- 221654
- Umair Mubashir- 221651

11. Conclusion

This Complex Engineering Activity provided practical exposure to industrial automation using PLCs. Through systematic analysis, logical design, and ladder logic implementation, safe and efficient automation solutions were developed for real-world scenarios. The activity strengthened understanding of sequencing, control strategies, and industrial safety, fulfilling the intended learning outcomes of the course.